

Omais Shafiq

PhD Researcher in Computational Structural & Earthquake Engineering

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Research Profile

PhD researcher specializing in AI-integrated seismic fragility assessment with **10 journal publications** (4 under review in *Structures* Q1, *MethodsX* Q1). Creator of the **EIDA Framework**—a high-throughput automated pipeline integrating C#, MATLAB, and ETABS API for parametric nonlinear analysis across thousands of structural configurations (50,000+ fragility curves generated). Expertise in transparent machine learning (XGBoost, ANN, SHAP) for code-ready seismic design equations.

Core Research Contributions

EIDA Framework (*under review in MethodsX Q1*) — Automated C# + MATLAB pipeline enabling parametric IDA at scale via ETABS API with direct text injection for bulk nonlinear hinge assignment, Hunt-and-Fill ground motion scaling, and per-archetype Gazetas spring recalculation for soil–structure interaction.

Transparent ML for Seismic Fragility (*under review in Structures Q1*) — XGBoost + ANN ensemble with Meta-Regression for Feature Importance Analysis converting complex IDA results into closed-form dimensionless fragility equations for spreadsheet evaluation.

Technical Expertise

Computational:	Incremental Dynamic Analysis, probabilistic seismic demand modeling, parametric nonlinear time-history analysis, fragility curve generation
Machine Learning:	XGBoost, ANN, meta-regression (GLME), transparent AI for structural engineering
Programming:	C# .NET, MATLAB, ETABS API automation, COM marshalling, fault-tolerant pipeline design
Engineering:	Performance-based earthquake engineering, soil–structure interaction, RC frame design, lumped plasticity modeling

Education & Status

PhD — Structural Engineering (<i>Viva: Sept 2026, Expected graduation: May 2027</i>)	UMPSA, Malaysia
Supervisor: Dr. Saffuan Bin Wan Ahmad • Thesis: AI-integrated seismic fragility assessment	
MSc — Structural Engineering (2020–2024)	UET Peshawar, Pakistan
BSc — Civil Engineering (2015–2019)	UET Peshawar, Pakistan

Selected Publications

Under Review (Q1 Venues)

- **Shafiq, O.**, et al. Factors Affecting Seismic Fragility of RC Grid-Like Frame Structures: A Parametric Meta-Regression Approach. *Structures* (Rev. Submitted—R1)
- **Shafiq, O.**, et al. Transparent ML Framework for Seismic Fragility Derivation: A Closed-Form Dimensionless Approach. *Structures* (Submitted)
- **Shafiq, O.**, et al. Parametric Sensitivity of the Seismic Demand Scaling Exponent to Design Variability and SSI in RC Bare Frames. *Structures* (With Editor)
- **Shafiq, O.**, et al. An Automated Extended IDA Pipeline for High-Throughput Fragility Data Generation with SSI in ETABS. *MethodsX* (Submitted)

Published / In Press

- **Shafiq, O.**, et al. (2025). Fragility Curves Development for Single Span Slab Bridges: A Case Study for Peshawar, Pakistan. *J. Adv. Research in Applied Mechanics*, 133(1), 177–190 [Scopus]
- Syed Mustorpha, S.I., et al. (2025). Advancements in Earthquake Research and Seismic Risk Analysis: An In-Depth Review Centered on Malaysia. *J. Adv. Research in Applied Sciences & Eng. Tech.*, 63(2) [Q2]
- Waqas, M., et al. (2025). Seismic Fragility Characterization of Grid-Like Frame Structures. *Construction*, 5(2) [Q3]
- Waqas, M., et al. (2025). Advances in Earthquake Mitigation in Malaysia: A Review. *Jurnal Teknologi* (In Press) [Q3]

Postdoc Interests

AI-driven structural design (generative models, graph neural networks), transparent ML for code-ready fragility formulations, physics-enhanced machine learning, regional seismic vulnerability assessment, performance-based earthquake engineering at scale. Open to joint research, co-authored publications, and grant proposals leading to postdoctoral positions starting mid-2027.